

# CHRISTOPHER BITLER

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## SENIOR SOFTWARE ENGINEER

*BUILDING RELIABLE APPLICATIONS, SCALABLE SYSTEMS, EFFICIENT ENGINEERING WORKFLOWS,  
AND PRACTICAL AI SOLUTIONS FOR COMPLEX BUSINESS PROBLEMS*

### PROFESSIONAL EXPERIENCE

**Opencash (Acquired by InvestView)** | Haverford, PA - Remote (September 2025–Present)

#### SOFTWARE ENGINEER

##### *Platform Architecture, Security & Reliability*

- Separated PII tokenization and job management from the main backend into dedicated NestJS/TypeScript microservices, improving separation of concerns and limiting unnecessary access to sensitive data.
- Researched whether to build SSN tokenization internally or use an outside provider, comparing infrastructure, engineering time, required functionality, and cost before recommending a third-party solution that met security and functional requirements while avoiding the time and expense of an internal build.
  - Built the service using TDD and deterministic tokenization so duplicate SSNs could still be detected, restricted de-tokenization to authorized staff, and recorded all activity in WORM-protected S3 audit logs; had the service running in staging within one week.
- Designed the auditability and result-generation controls for Opencash's weekly promotional rewards process, allowing results to be reproduced and verified as unaltered during regulatory or audit review.
  - Implemented tamper-evident audit records in WORM-protected Amazon S3 and independently verifiable randomization using AWS KMS and multiple external entropy sources, including NIST and DRand, strengthening traceability and integrity controls across the process.
- Reworked scheduled-job processing after finding that EventBridge could schedule jobs but could not adequately retry failures or stop later jobs when an earlier step failed.
  - Built an AWS Step Functions/ECS workflow with standardized success/failure reporting and exponential-backoff retries.
  - Used the same framework for the roughly eight-job weekly scratch-off process, allowing processing to stop on a critical failure, alert the on-call engineer, and resume from the failed step after the problem was corrected; significantly reduced incomplete jobs.

##### *Integrations, Delivery & AI*

- Implemented automated reconciliation and monitoring for missing and anomalous data between OpenCash and its clearing firm using webhooks and scheduled SFTP feeds, with error handling around trade-order processing and updates to improve reliability and keep internal records aligned with clearing-firm data.
- Removed a manual TypeORM database-migration step that could be missed, run against the wrong environment, or leave the application and database out of sync.
  - Used AWS CodeBuild as a GitHub Actions runner inside the VPC to run migrations automatically during staging and production deployments.
- Built an OpenAI API-based support chatbot so users could get answers about the application and general trading terminology without contacting support.
  - Compared available AI options and used OpenAI Structured Outputs, Guardrails, and prompt instructions to keep the chatbot from providing financial advice; reduced support calls and emails.
- Simplified a poorly documented React Native/Expo release process that was prone to mistakes by creating a GitHub Action that generated a pull request showing all changes since the prior release.
  - Once approved and merged, the workflow automatically started the Expo release, reducing release errors and making the process much easier for newer engineers to learn and run.

**Intuit** | Mountain View, CA (May 2020–April 2025)

**SOFTWARE ENGINEER 2 (MAY 2021–APRIL 2025) / SOFTWARE ENGINEER (MAY 2020–MAY 2021)**

*Earned 22 peer-nominated Spotlight Awards over four years for engineering contributions across lending, payments, and cryptocurrency products within QuickBooks Capital and QuickBooks Online.*

**Product, AI & Full-Stack Development**

- Transformed QuickBooks Capital’s underwriting process by implementing a TypeScript/React generative AI solution that analyzed loan applications, identified issues, and provided remediation steps, reducing review time and allowing underwriters to process more applications without sacrificing review quality.
- Developed an AI-powered Node.js/TypeScript migration tool that automatically converted JavaScript to TypeScript, eliminating manual conversion work and saving engineering time; one of four teams recognized with an organization-level award for engineering efficiency.
- Led the UI implementation for QuickBooks Crypto’s buy/sell/hold feature, using TypeScript, React, React Query, and Kotlin to give customers the ability to purchase, hold, or sell cryptocurrency through custodial wallet partners using QuickBooks Checking funds.
- Redesigned portions of the QuickBooks Line of Credit acceptance flow in TypeScript/React, improving input error handling and reducing support tickets while supporting efforts to increase loan approvals and volume.

**Architecture, APIs & Developer Productivity**

- Reduced production issues from breaking API changes by developing a Groovy/Jenkins build pipeline that automatically generated TypeScript definitions from OpenAPI schemas; earned a Spotlight Award for the solution and for presenting it to the development team.
- Connected QuickBooks UI flows with multiple payment and lending microservices through Apollo GraphQL, allowing user, payment, and loan data to be queried and updated across services.
- Increased line-of-credit signups among customers who declined term-loan offers by implementing Java/Spring Boot backend logic that triggered targeted marketing emails through Camunda.

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## CO-OP EXPERIENCE

**Intuit** | Mountain View, CA – Software Engineer Intern (May 2019–December 2019)

**Datto** | Rochester, NY – Software Engineering Intern (January 2017–August 2017 and May 2018–August 2018)

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## TECHNICAL SKILLS / AREAS OF EXPERTISE

**Languages:** TypeScript | JavaScript | Java | Kotlin | HTML/CSS

**Frontend:** React | React Native | React-Query | Redux | Next.js | Tailwind CSS | Apollo Client (GraphQL)

**Backend & APIs:** Node.js | NestJS | Spring Boot | Apollo Server (GraphQL) | REST | OpenAPI | Microservices | TypeORM

**Cloud, Containers & CI/CD:** GitHub Actions | Jenkins | Terraform | Docker | Kubernetes

AWS (ECR, ECS, S3, Step Functions, CodeBuild, Cognito, IAM, CloudWatch, CloudFront, Secrets Manager)

**Data, Testing & Observability:** PostgreSQL | MySQL | MongoDB | TDD | Jest | Playwright | React Testing Library |

Grafana/Prometheus | Splunk | OpenSearch

**Engineering Practices & Architecture:** System Architecture | API Design | Security Engineering | Performance Optimization | Developer Tooling & Automation

**AI Engineering:** Generative AI | OpenAI API | Structured Outputs | Guardrails | OpenAI Codex

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## EDUCATION / CERTIFICATIONS

**B.S. Computer Science, Rochester Institute of Technology**

**Developing Generative AI Applications with Python** | IBM | Certificate ID: [09a32664e748450c9c6e28dff07071db](#)

**Verified Certificate for Introduction to Generative AI** | IBM | Certificate ID: [c47f5313853c4ca9bd0463e05b38258e](#)